

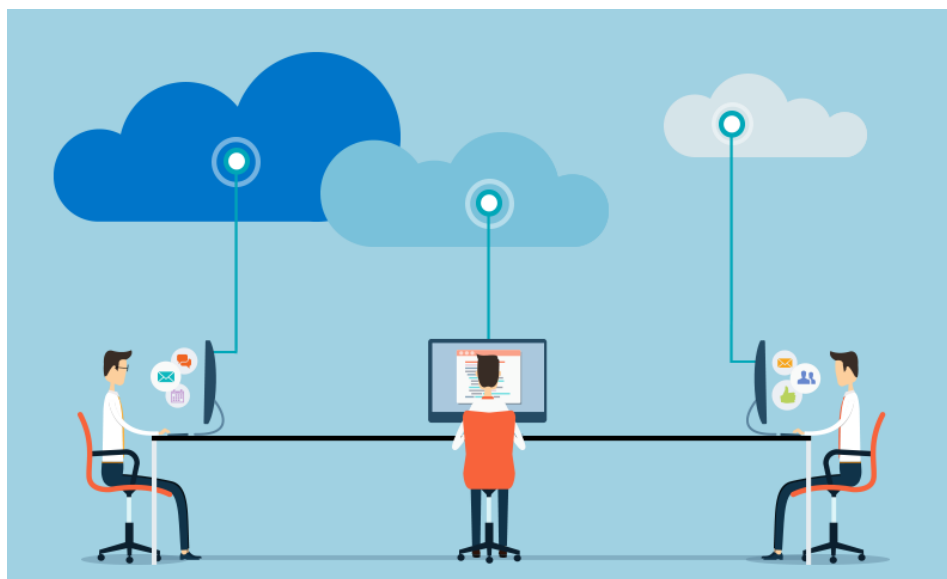


TECNOLÓGICO
NACIONAL DE MÉXICO



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Departamento de Ingeniería en Sistemas Computacionales



Materia: Computo en la Nube

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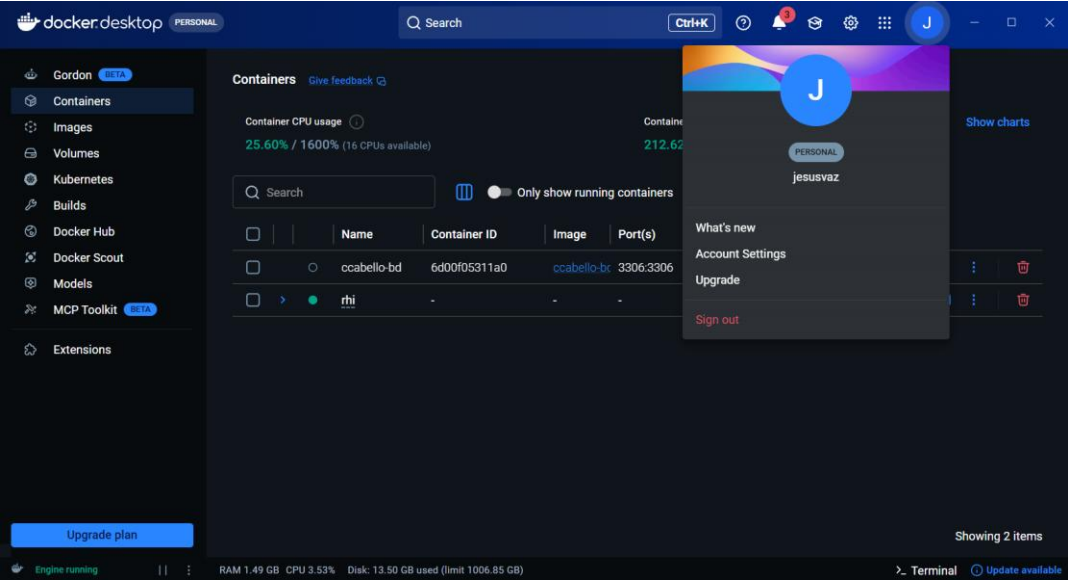
(7 – 8 hrs)

**“Actividad 5.4 Instalación de la plataforma de código abierto
Docker”**

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```
Windows PowerShell
PS C:\Users\jesus> docker --version
Docker version 29.3.1, build c2be9cc
PS C:\Users\jesus> |
```



```
Windows PowerShell
PS C:\Users\jesus> docker --version
Docker version 29.3.1, build c2be9cc
PS C:\Users\jesus> docker images

IMAGE                ID                DISK USAGE    CONTENT SIZE  EXTRA
ccabello-bd:latest   082a667d97e9     797MB         0B            U
mysql:latest         85bdb7e77f7f     920MB         0B            U
node:18              b50882bc3670     1.09GB        0B            U
node:20              d19e81beed0f     1.1GB         0B            U
postgres:15-alpine  cd048ee12a8e     274MB         0B            U
rhi-backend:latest  b8bce1601815     145MB         0B            U
rhi-frontend:latest 0312bef52483     271MB         0B            U
PS C:\Users\jesus>
```

```
Windows PowerShell
PS C:\Users\jesus> docker run -d -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
3531af2bc2a9: Pull complete
ce776bbcd8d: Pull complete
85c66128325a: Pull complete
4677c2a9a3d4: Pull complete
ff048f1f2159: Pull complete
677c63196868: Pull complete
801a1ad15b4e: Pull complete
Digest: sha256:3acc8b9cc6421db48299559e937b2ade69ddd609453b4f44c21ccb239d614535
Status: Downloaded newer image for nginx:latest
05ec068d54a4dd2233a86e0610be119f44989dabb7359c04290639afbc82b67e
docker: Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8080 -> 127.0.0.1:0: listen tcp 0.0.0.0:8080: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.

Run 'docker run --help' for more information
PS C:\Users\jesus> docker ps -a

CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS
05ec068d54a4   nginx               "/docker-entrypoint..." 12 seconds ago Created
180e9b258dfff elegant_kapitsa     "docker-entrypoint.s..." 20 hours ago   Up 14 minutes   0.0.0.0:5173->5173/tcp, [::]:5173->5173/tcp
858a82b1453b   rhi-frontend       "docker-entrypoint.s..." 20 hours ago   Up 14 minutes   0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp
001->3001/tcp   rhi-backend        "docker-entrypoint.s..." 20 hours ago   Up 14 minutes (healthy) 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
b8ae9a1fbc57   postgres:15-alpine rhi_db                  7 months ago   Exited (0) 6 months ago
432->5432/tcp   ccabello-bd        "docker-entrypoint.s..."
6d00f05311a0   ccabello-bd        "docker-entrypoint.s..."
PS C:\Users\jesus> |
```

```

PS C:\Users\jesus> docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

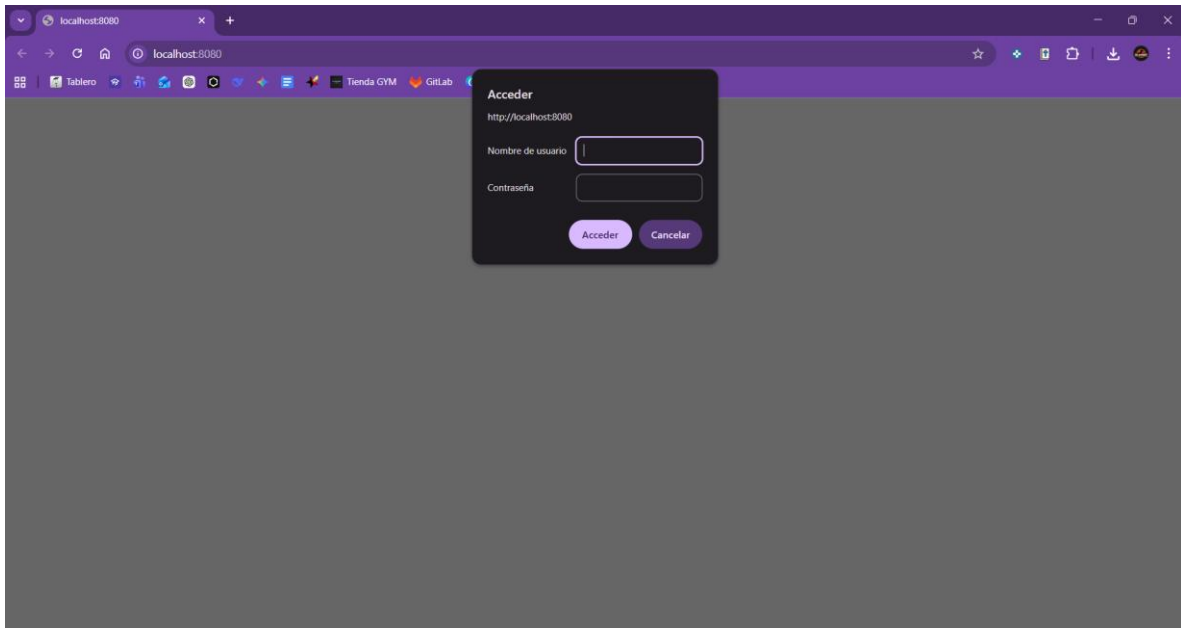
PS C:\Users\jesus>

```

The screenshot shows the Docker Desktop application window. The left sidebar contains navigation options: Gordon (BETA), Containers (selected), Images, Volumes, Kubernetes, Builds, Docker Hub, Docker Scout, Models, MCP Toolkit (BETA), and Extensions. The main area is titled 'Containers' and shows system metrics: Container CPU usage at 25.03% / 1600% (16 CPUs available) and Container memory usage at 212.79MB / 7.25GB. A search bar and a toggle for 'Only show running containers' are present. Below is a table of running containers:

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	ccabello-bd	6d00f05311a0	ccabello-bc	3306:3306	0%	7 months ago	
☐	elegant_kapitsa	05ec068d54a4	nginx	8080:80	0%		
☐	interesting_jeps	821a22c1403b	hello-world		0%	2 minutes ago	
☒	rhi	-	-	-	25.03%	17 minutes ago	

At the bottom, a status bar shows 'Engine running', 'RAM 1.46 GB', 'CPU 5.73%', 'Disk: 13.67 GB used (limit 1006.85 GB)', and buttons for 'Terminal' and 'Update available'. A 'Showing 4 items' indicator is also present.



Preguntas que surgieron durante la instalación o la ejecución de las pruebas:

1. ¿Por qué el comando docker run a veces tarda mucho en la primera ejecución y luego es instantáneo?

Descubrí que la primera vez Docker debe descargar (*pull*) todas las capas de la imagen desde Docker Hub y una vez que la imagen está en el almacenamiento local, Docker solo necesita instanciar el contenedor.

2. ¿Por qué se presentan errores de "Access Denied" al ejecutar comandos de Docker en Windows a pesar de tener la aplicación abierta?

El error se debe a la falta de permisos del usuario para interactuar con el motor de Docker, pero se resolvió ejecutando la terminal como Administrador y agregando el usuario al grupo local "docker-users", aparte se verificó que el firewall de Windows no bloqueara el backend de WSL2, garantizando la comunicación correcta con el demonio de Docker.

3. ¿Por qué el comando docker --version no es reconocido en la terminal a pesar de haber finalizado la instalación?

El problema se debe a que la ruta de los ejecutables de Docker no se agregó correctamente a las Variables de Entorno (PATH) de Windows o a que el motor de Docker Desktop no se ha iniciado, pero se resolvió agregando manualmente la ruta C:\Program Files\Docker\Docker\resources\bin al PATH del sistema y reiniciando la terminal.